

Mild dehydration impairs cognitive performance and mood of men.

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The present study assessed the effects of mild dehydration on cognitive performance and mood of young males. A total of twenty-six men (age 20.0 (sd 0.3) years) participated in three randomised, single-blind, repeated-measures trials: exercise-induced dehydration plus a diuretic (DD; 40 mg furosemide); exercise-induced dehydration plus placebo containing no diuretic (DN); exercise while maintaining euhydration plus placebo (EU; control condition). Each trial included three 40 min treadmill walks at 5.6 km/h, 5 % grade in a 27.7°C environment. A comprehensive computerised six-task cognitive test battery, the profile of mood states questionnaire and the symptom questionnaire (headache, concentration and task difficulty) were administered during each trial. Paired t tests compared the DD and DN trials resulting in >1 % body mass loss (mean 1.59 (sd 0.42) %) with the volunteer's EU trial (0.01 (sd 0.03) %). Dehydration degraded specific aspects of cognitive performance: errors increased on visual vigilance ($P = 0.048$) and visual working memory response latency slowed ($P = 0.021$). Fatigue and tension/anxiety increased due to dehydration at rest ($P = 0.040$ and 0.029) and fatigue during exercise ($P = 0.026$). Plasma osmolality increased due to dehydration ($P < 0.001$) but resting gastrointestinal temperature was not altered ($P = 0.238$). In conclusion, mild dehydration without hyperthermia in men induced adverse changes in vigilance and working memory, and increased tension/anxiety and fatigue.